```
#Date: 2026-08-29
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#Problem 2 of assignment

.data
A: .word 0
B: .word 1

.globl main
.text
main:

#Half Adder for 2 1 bit numbers
#SUM = A XOR B
#CARRY = A AND B
lw $t0, A
lw $t1, B
and $s2, $t0, $t1 #This is carry but

#XOR = (A.!B) + (!A.B)
nor $t2, $t0, $t0 #!A
nor $t3, $t1, $t1 #!B
and $t4, $t0, $t3
and $t5, $t1, $t2
or $s1, $t5, $t4 #This is sum bit

#Half Subtractor (we are calculating A-B)
#$s1 also stores the difference since sum and difference are the same

#Instead of carry, we should calculate borrow
#Truth Table for subtractor
#A  B   A-B(difference, borrow)
#0  0   (0,0)
#0  1   (1,1)
#1  0   (1,0)
#1  1   (0,0)

#Borrow = !A.B
and $s3, $t2, $t1



li $v0, 10 # Sets $v0 to "10" to select exit syscall
syscall
```